# TIC TAC TOE ARENA

by

BHUVISHA GOHIL

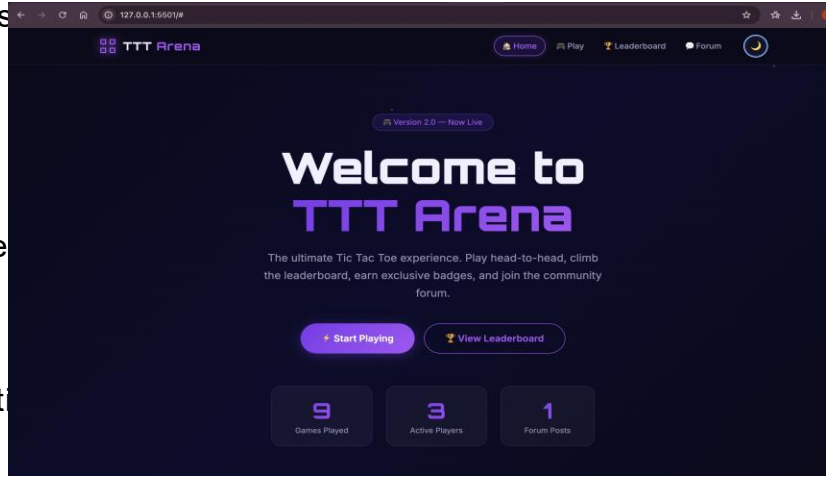
150096725190

# 1. Problem Statement

Most existing online Tic Tac Toe games are very simple and only provide the basic gameplay experience. They generally lack advanced features that make the game more engaging for users. Some common limitations include:

- No score tracking system across multiple matches
- Lack of player progression or ranking system
- No customization options such as themes or UI settings
- Absence of community interaction features like forums
- No motivation for players to continue playing after one game

These limitations reduce user engagement and long-term interaction with the platform.




**Objective of the Project**

The objective of this project is to design and develop an interacti Tic Tac Toe web application that enhances the traditional gameplay by including additional features such as:

- Real-time score tracking
- Badge-based ranking system
- Leaderboard for player comparison
- Theme customization
- Forum for discussion and strategy sharing

# 2. Solution

To solve the limitations of traditional Tic Tac Toe games, the project introduces **TTT Arena**, an advanced browser-based game platform.
TTT Arena is designed to provide a **more engaging and interactive user experience** while maintaining the simplicity of the original game.
Key features of the system include:

**1. Interactive Game Board**

- A 3×3 grid where players take turns placing X and O.

**2. Scoreboard System**

- Tracks wins, losses, and draws for each player.

**3. Badge and Ranking System**

- Players earn badges based on their performance and number of wins.

**4. Leaderboard**

- Displays top players ranked according to wins and win rate.

**5. Community Forum**

- Allows players to share strategies, ask questions, and interact with other users.

**6. Theme Customization**

- Users can switch between **Dark, Neon, and Light themes**.

The platform is designed to work **entirely within the browser**, without requiring server-side integration.

# 3. Working/Methods & Algorithms.

The game logic is implemented using **JavaScript algorithms** that control gameplay and determine the outcome of each match.

**Game Board Representation**

- The game board is represented using a **9-element array**.
- Each element corresponds to a cell in the 3×3 grid.
- The value can be **X, O, or empty**.

**Winning Algorithm**

After each move, the system checks predefined winning patterns:

Rows:

- [0,1,2]
- [3,4,5]
- [6,7,8]

Columns:

- [0,3,6]
- [1,4,7]
- [2,5,8]

Diagonals:

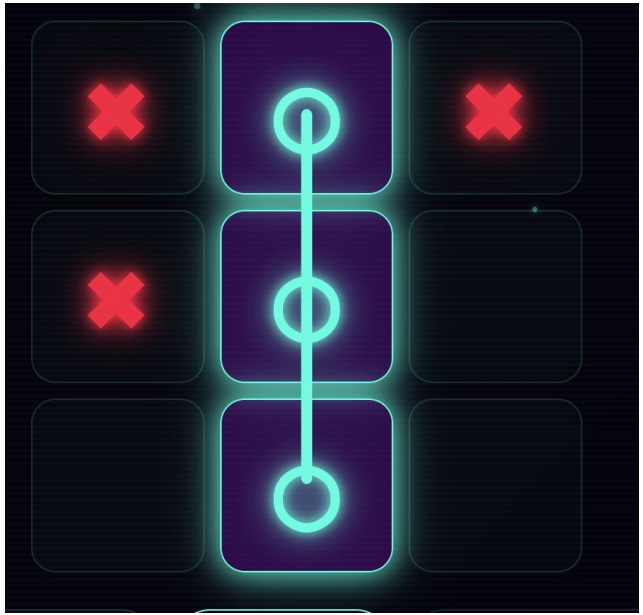
- [0,4,8]
- [2,4,6]

If any pattern contains the same symbol (X or O), the system declares a winner.

If all cells are filled without a winning pattern, the game ends in a **draw**.

**Additional Algorithms**

- Badge assignment based on total wins
- Leaderboard ranking using win statistics
- Forum post management using **localStorage**



# 4. Conclusion

The TTT Arena project demonstrates how a simple traditional game can be enhanced using modern web development technologies.

The platform successfully integrates gameplay with interactive features such as:

- Real-time score tracking
- Leaderboard and badge system
- Theme customization
- Community discussion forum

These features significantly improve user engagement and overall gaming experience.

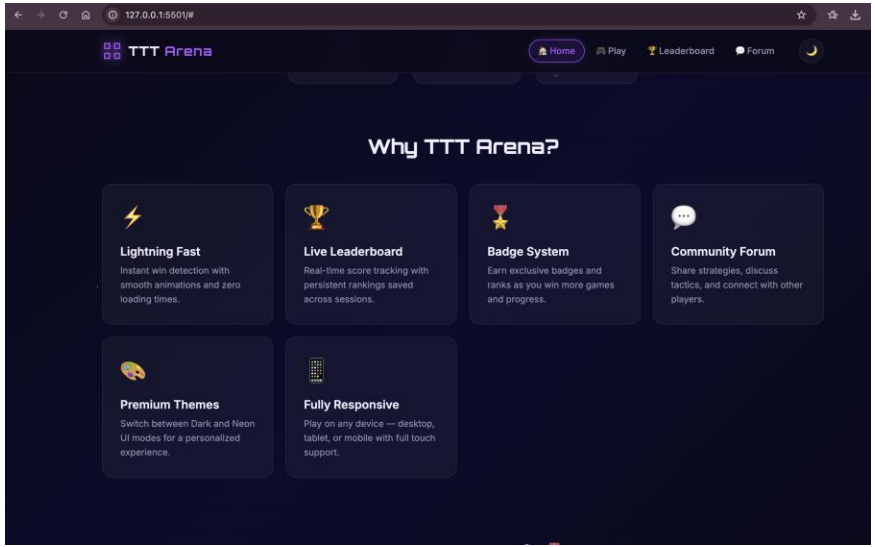
The project also demonstrates practical implementation of:

- HTML5 for structure
- CSS3 for styling and animations
- JavaScript for game logic and interactivity
- LocalStorage for persistent data storage

Future Enhancements

The system can be further improved by adding:

- Artificial Intelligence (AI) opponent
- Sound effects and background music
- Online multiplayer support
- User authentication for multi-device progress



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# Thank You